

Project Summary

The group unanimously selected the SSH Grocery Delivery Coordination EDR proposed by Mohammed Alomari because it was the most realistically doable idea within the tight timeline. Every group member reviewed the proposed EDR and Mohammed gave a quick explanation of it, and everyone agreed to use it. For deployment and testing, we used Ngrok to easily share and expose the application.

We developed the following key features:

- **Shared Cart Management:** Flatmates could collaboratively add, update, and track grocery items.
- **Cost Splitting:** Automatically calculated and split costs among users.
- **Notification System:** Sent alerts for cart updates, new items, or purchases.
- **Secure Data Handling:** Django's CSRF protection and access controls ensured data security.

On the frontend, we used HTML, CSS, and JavaScript to create a responsive and user-friendly interface that seamlessly integrated with the Django backend. The interface allowed easy navigation and clear visualization of grocery lists and costs. We also used Docker for containerisation which greatly improved efficiency and ensured consistent deployment across different environments. We built the backend with Django, managed data with .json files, and used GitHub for version control. Regular team meetings and clear communication kept us organised, enabling us to deliver a functional and efficient prototype.

Project Report

What we achieved

Our team developed a prototype to simplify grocery coordination for flatmates, guided by our Engineering Design Requirements (EDR). Focusing on core features—shared cart management, automated cost splitting, and notifications—we demonstrated how our design enhances convenience, transparency, and cost efficiency.

How we achieved it

We used Django for the backend and standard web technologies (HTML, CSS, JavaScript) for the frontend, ensuring a responsive interface and reliable functionality. Strategic project planning involved dividing tasks into frontend, backend, database management, containerisation, and continuous integration, assigning roles based on strengths.

Roles and Responsibilities

- **Backend and Frontend Development:** Mohammed AlOmari focused on Building the Backend, while also contributing to Front-end development.
- **Project Management and Code Quality:** Sattam Almuwallad managed the project, coordinated tasks, reviewed code, and fixed bugs in both the frontend and backend.
- **Version Control and Database Management:** Yasser Basha handled GitHub workflows and managed the database.
- **Backend and Deployment:** Mohammed Aljowaie took charge of backend development and deployment tasks.

Project Management

We used GitHub Issues to track tasks and held regular Discord meetings to address challenges promptly. When switching from SQLite to PostgreSQL, effective communication ensured smooth adaptation without losing momentum. This collaboration maintained a steady workflow and successful outcomes.

Software Engineering Techniques

During our project, we really focused on following good software engineering Techniques. We used GitHub extensively for version control, setting up separate branches for each new feature we worked on to keep things organized and manageable. When it was time to merge those features into the main codebase, we used pull requests and made sure to do thorough code reviews to keep our code quality high. We also set up branch protection rules and merged our changes regularly, which helped us catch any problems early and stay on track throughout the project.

Testing was a big part of how we worked. We started out with Django's default testing framework (tests.py) but eventually switched to Pytest because it gave us more flexibility for writing tests. We focused on writing unit and integration tests, paying special attention to edge cases and the core features of our system. Automated tests, powered by GitHub Actions CI workflows, gave us instant feedback whenever we made changes. This setup not only prevented buggy code from being merged but also fit really well with our Agile and Extreme Programming methodologies, allowing us to release small, tested updates quickly.

We also set up continuous deployment. Whenever our tests passed, we could roll out new features in small, frequent updates. This kept our codebase stable while still letting us make steady improvements to the prototype. Using Docker and docker-compose for containerization was a game-changer—it made it easy to create consistent environments, both for development and deployment. Learning how to use Docker took

some time, but it really paid off, especially when we had to make changes to database configurations.

For managing dependencies, we used Python virtual environments and a requirements.txt file. This kept our project's requirements isolated and avoided conflicts. At first, we thought about using PostgreSQL, but for the prototype, we decided to stick with Django's built-in database and .json files to keep things simple. This let us focus on getting the core features working and made the system more stable. When we did experiment with PostgreSQL, Docker made the process much smoother by ensuring consistent setups across different machines and operating systems, which was especially helpful during database migrations.

Technical Accomplishment and Evaluation

Our prototype successfully brought the planned features to life, meeting all the objectives outlined in the EDR. The shared cart management feature enabled multiple users to collaboratively add and update grocery items with ease. Automated and transparent cost splitting ensured fairness, while the notification system kept everyone in the loop about updates and purchases. These features came together seamlessly to create a functional and practical solution. By testing these functionalities against our requirements, we confirmed the design's viability. Challenges, like ensuring seamless frontend-backend data handling, were resolved by aligning form fields and adding debug statements. This careful troubleshooting built confidence in our solution's technical soundness.

To simulate external systems, we created mock data rather than relying on live services. Django's framework handled backend logic, and the frontend rendered these mock datasets, enabling us to validate core behaviors without external dependencies.

Reflections and Lessons Learned

Throughout the project, we honed key technical and teamwork skills. We improved our understanding of Agile methods, version control, CI/CD, Docker, and Django. Our communication skills also strengthened: regular discussions, clear documentation, and prompt issue tracking were critical to our success. These lessons have prepared us for tackling more complex projects in the future.

Peer reflections

Mohammed Aljowaie Reflections

Reflection on Mohammed Alomari

Mohammed was an important member of our team, especially in the initial stages of the project. He took the lead in developing the basic architecture for both the frontend and backend, which provided a solid foundation for our work thereafter. His early contributions paved a clear direction where we could thrive upon. Without his foundational work, the advancement of the team would have been much slower and more chaotic.

One of Mohammed's standout characteristics is his flexibility. He seamlessly shifted between frontend and backend tasks as needed and proved to be very adaptable. This flexibility allowed us to tackle the challenges we faced, especially in those critical points in the project. Mohammed's pragmatic mindset and focus on simplicity ensured we did not overcomplicate the prototype. Many times, he followed a very conservative approach and avoided big-feature ideas, which felt limiting at times. Nevertheless, this pragmatic focus kept us grounded and allowed us to complete the project within our predefined constraints. Mohammed's approach was especially helpful in keeping the project viable. At times, he would discard ideas that were a bit more creative or less certain; however, in hindsight, this prevented us from potential pitfalls. For the future, I believe Mohammed can further increase his influence by coupling his pragmatic nature with an acceptance for exploring

options. In all, Mohammed's dedication, flexibility, and valuable contributions were significant to our success. His contributions kept the project on course, and I truly appreciated his commitment, flexibility, and collaborative spirit throughout the project.

Reflection on Yasser Basha

Yasser became an essential member of our team, especially when it came to explaining the Git workflows. Before his mentoring, I struggled with the commands and versioning in Git. Many times, I cloned the repository and pushed right to the main branch, which created conflicts on many occasions. Yasser explained all the basic usage of Git,

including creating branches, making a pull request, and merging changes in steps. His help greatly increased my productivity and allowed the team to avoid common pains.

Beyond his technical competencies, Yasser was an exceptional timekeeper who ensured timeliness and adherence to deadlines. He always attended meetings in time and completed his tasks on time, which helped keep the team organized and focused. His reliability instilled confidence in our ability to make constant progress. Also, Yasser valued being responsible for his work and wanted to be consulted before any changes were made to his code. Although this might have seemed strict, it ultimately kept our codebase intact and reduced bugs. Yasser's good communication and systematic approach were of great importance for team harmony. He made sure every member knew what they needed to do and followed best practices. His attention to quality and organization influenced the entire team.

Looking ahead, Yasser's technical skills, his sense of responsibility, and collaborative spirit will be continued sources of enhancement to productivity. His contribution was invaluable to the success of our project, and I indeed valued his support, guidance, and attention to detail. His skills and work ethics make him a great colleague to rely on for future projects

Reflection on Sattam Almuwalad

Sattam proved to be a reliable and supportive team member, contributing much with his commitment and positive attitude. At the beginning of the project, he faced some problems with GitHub workflows, but he did not give up. When technical difficulties barred him from pushing code, he concentrated on project management and coordination. This flexibility ensured that the team had a well-organized and productive workflow.

Sattam took charge of scheduling meetings, facilitating peer code reviews, and ensuring that everyone was aligned on tasks. His organizational skills helped us stay on track and address challenges efficiently. During backend development, he often spent hours reviewing my work, offering feedback, and handling smaller tasks to lighten my workload. His commitment to supporting the team in any way possible was truly appreciated.

One of the best examples that come to mind regarding Sattam's dedication is when we were stuck with a difficult bug and couldn't fix it despite working for hours. Exhausted, we decided to stop work for the day. To my surprise, the next morning Sattam came with the issue fixed. His determination and willingness to go out of his way made a real difference.

At the end of this project, Sattam was much more technically capable. If he continues to develop his skills in GitHub, he will become an even stronger contributor. His positivity, flexibility, and commitment were indispensable to our group. I truly valued his input, help, and collaboration throughout the project.

Sattam Almuwallad Reflections

Reflection on Mohammed Aljowaie

Mohammed Aljowaie was an excellent teammate to work with. He didn't just get the job done — he went above and beyond when things got tough. When Docker was assigned to someone else who ran into difficulties, Mohammed immediately said, "Let me try." He spent hours troubleshooting and testing different solutions until he got it working. Thanks to his persistence and hard work, we successfully containerized the project, making deployment

and efficient. That said, Mohammed had some challenges with time management. Missing an important meeting early on delayed the implementation of some of his ideas. To his credit, he recognized the issue and put in extra hours to catch up, showing real commitment to the project and the team.

He also occasionally struggled with GitHub conflicts, sometimes leaving them for others to resolve. With more practice managing version control, I know this is something he can improve.

Mohammed's dedication, adaptability, and technical skills make him a reliable teammate. With better time management and a stronger grasp of GitHub workflows, his contributions will be even more impactful. I'm looking forward to working with him again and seeing how he continues to grow

Reflection on Mohammed AlOmari

Working with Mohammed AlOmari was a great experience. I learned so much from his frontend development skills, and he brought tremendous value to the project. His work gave the project a professional and user-friendly feel that made the user experience enjoyable.

I'll never forget the time we couldn't agree on a design, and out of nowhere, Mohammed came up with a solution that satisfied everyone. His problem-solving skills and collaborative approach really helped us move forward when we were stuck.

Mohammed's versatility was a huge asset to the team. Whether it was backend development or database tasks, he was always ready to jump in and help. That kind of

willingness to support the team showed how reliable he is. There was a small hiccup at the beginning when he pushed code without checking with the team or using pull requests, which caused some confusion. But to his credit, he quickly realized his mistake and helped us set up a better workflow. This improved our collaboration and made managing the code much smoother.

Moving forward, I'd love to see Mohammed continue developing his backend skills, which would make his contributions even stronger. He's adaptive, creative, and a great team player. I'm genuinely looking forward to working with him again and seeing what he brings to future projects.

Reflection on Yasser Basha

Yasser Basha has been a tremendous asset to our team. His deep knowledge of database management and GitHub workflows was invaluable. He walked us through best practices for branching, pull requests, and rebasing, keeping the development process smooth and organized. His clear explanations made everyone feel confident managing their code, which helped us a lot.

Yasser's teamwork skills really stood out. Whenever someone had a problem with their code, he was always there to help and keep things moving. Early on, he underestimated how long it would take for everyone to get comfortable with GitHub, which caused a few delays. But he made up for it by offering ongoing support and one-on-one help, making sure no one was left behind.

For future projects, setting more realistic timelines for training and onboarding would help things run even smoother. Yasser's problem-solving skills, patience, and adaptability are a huge asset to any team. I truly appreciate his effort and the positive energy he brought to our work. I'm confident he'll keep excelling, and I look forward to working with him again.

Mohammed Alomari Reflections

Sattam Almuwallad

Working with Sattam felt like working with somebody who cared a lot about keeping us on the same page. First, he mostly concentrated on the front-end, maintaining the look and feel of our prototype, while clearing those small, pesky bugs. But what really stood out to me was how he stepped up in managing our workflow. He took up the role of managing our meeting schedules and code reviews-making sure each one of us knew

what happened after, so no one gets lost in the way. That kind of organization really saved us from a lot of headaches when we had to juggle several things at once.

Sattam did hit a rough patch early on. He wasn't that accustomed to GitHub and how he could properly push or pull code. For example, we spent two meetings to stop and fix some version control issues. Honestly, what really impressed me most was his response. Instead of getting frustrated or giving up, he asked questions and tried to learn. Through consistent practice and a willingness to learn, he became proficient in managing merges and resolving conflicts. His growth in this area was both impressive and inspiring to the rest of the team.

Beyond his technical contributions, Sattam fostered a positive and inclusive team environment. He encouraged open communication, respected everyone's input, and celebrated our collective growth as developers. Reflecting on his role, Sattam expressed a desire to have embraced project management earlier—a field in which he clearly excels. His blend of technical effort, leadership, and positivity significantly contributed to our team's success and morale.

Yasser Basha

Yasser played a huge part in keeping our project on track and holding us to a higher standard. Since he has previous experience working with projects, he made the whole journey a lot easier, by guiding us through proper software engineering practices. He had a solid grip on the “right” way to do things, like how to organize our code, use proper workflows, and maintain clear documentation. When he introduced GitHub Actions for continuous integration and testing, as well as branch protection rules, it felt like we were moving from a casual coding group to a real development team. With every pull request under scrutiny, we spent less time worrying about messy merges and more time building features with confidence.

At first, a lot of us (myself included) struggled to wrap our heads around GitHub's workflow. Yasser underestimated how long it would take us to get comfortable, and that led to some early bumps. But instead of shrugging it off, he adapted by breaking things down into simpler steps, offering extra guidance, and patiently answering our questions. It was clear he wanted everyone to succeed, not just finish the project.

Over time, the things Yasser taught us finally sunk in. He supported communication, made sure tasks were properly assigned and understood, and insisted on testing before merging anything significant. That consistency and support helped us improve both as individuals and as a team. Because of Yasser, we not only finished a project but also learned how to build something the right way together.

Mohammed Aljowaie

Working with Mohammed Aljowaie was quite an enriching experience. He has proved that the usage of appropriate deployment strategies, any product can turn into a winning one. Throughout our collaboration, Mohammed took on several very critical responsibilities that would keep our project running, as well as saving a huge amount of time and effort down the line.

From the very beginning, he focused on backend functionalities: implementing detailed logging and search features that made it much easier to keep track of system activities and quickly retrieve relevant information. By maintaining these logs and refining the search capabilities, he ensured both developers and users could understand the system's behaviour and troubleshoot issues without getting lost in guesswork. His attention to all these little details avoids turning small bugs into big headaches later.

Beyond the backend logic, Mohammed was instrumental in containerizing our application using Docker. This allowed the whole team to work in consistent environments and avoid the "it works on my machine" problem. He also facilitated deployment through ngrok, allowing others to access our in-development application much more easily. With his guidance, the once cumbersome process of sharing and testing new features became almost effortless.

What impressed most about Mohammed was his attitude toward researching new tools and refining processes. The bumps along the way were there, but rather than frustration, he approached them with curiosity and persistence. By the end of the project, his technical contributions and problem-solving mindset significantly shaped our application's stability and usability, setting a high standard for future projects.

Yasser Basha Reflections

Mohammed Aljowaie

Mohammed Aljowaie was a key part of our backend development team, working closely with Mohammed Alomari. His technical know-how was evident in the way he implemented containerization by using docker, which significantly improved the project's scalability. By implementing Docker, Mohammed helped us establish a more robust and future-proof deployment process. However, with these improvements came some challenges in communication. I recall one instance where I tried running the

usual python manage.py makemigrations command, unaware that now with docker implemented we had to use different commands to run the project through the container. after 30 minutes of trying to figure out what was wrong in the code, I asked Mohammed about my issues running the project after which he explained that to explain the new command sequence, which made me realize how essential it is to communicate major changes up front.

I admired Mohammed's proactive attitude and his determination to learn from challenges. Even when things didn't go as planned, he stayed committed to improving his processes and keeping the codebase stable.

That said, I think he could take teamwork to the next level by sharing updates more consistently. Something as simple as a quick walkthrough or a note about major changes could prevent confusion and save everyone a lot of time.

Mohammed's technical skills and dedication played a vital role in the success of our project. With a little more focus on communication, I'm confident he'll continue to grow and lead complex backend projects with ease and expertise.

Sattam Almuwallad

Sattam Almuwallad was the team member responsible for the front-end. From the very beginning, he showed a proactive attitude, tackling bugs and issues with great attention to detail. One standout example was when he fixed an issue with the SSH (Student Smart Homes) logo scaling improperly. While the rest of us were focused on backend and database challenges, Sattam made sure these smaller, yet impactful issues didn't slip through the cracks.

What made Sattam's work particularly impressive was his ability to address cross-functional issues that affected both the front-end and back-end. He made sure that the front-end was not only visually appealing but also logically consistent with the other aspects of the project.

At the beginning of the project, He encountered some issues with GitHub, such as difficulties pushing changes to branches. Although this temporarily slowed his progress, he was able to quickly adapt and became one of the team's most dependable reviewers for pull requests. He would review pull requests the most of any members. His feedback was always thoughtful and thorough, often catching errors that others overlooked.

looking into the future, Sattam could enhance his knowledge and expertise by exploring advanced front-end frameworks and tools to broaden his skill set such as React and Node JS.

Sattam's dedication, adaptability, and willingness to support the team made him an invaluable teammate. I'm confident he'll continue to grow and make even greater contributions to future projects.

Mohammed Alomari

Mohammed Alomari was an outstanding teammate who excelled at solving complex technical problems. One example that stands out was a database issue where duplicate grocery items appeared as separate entries. I had been stuck on this problem for what felt like forever, but Mohammed suggested using a set to handle the duplicates. It was such a simple yet effective solution, and it completely resolved the issue while saving us a great deal of time. This example perfectly demonstrates his ability to think critically and offer practical solutions.

Mohammed also managed his responsibilities effectively, balancing his workload with Mohammed aljowaie in a seamless manner. While Mohammed aljowaie focused on implementing Docker and preparing for future scalability, Mohammed ensured the backend remained stable and could integrate smoothly with the front-end and database. His ability to connect the different components of the project was invaluable and helped the team stay aligned. Beyond his own tasks, he was always ready to assist others, whether by debugging tricky issues or helping Sattam with front-end logic.

One area where Mohammed could improve is in providing regular updates to the team. Sometimes it wasn't clear what he was working on, and a quick check-in would have made collaboration more efficient. Additionally, creating more detailed documentation of his work would help the team better understand and build on his contributions.

Overall, Mohammed's technical expertise and collaborative nature were key to our project's success. I look forward to seeing him continue to excel and take on even bigger challenges in the future.